

Silicone Elastomer with Self-Generating Zwitterions for Antifouling Coatings

Peng Hu, Haohang Zeng, Huan Zhou, Cong Zhang, Qingyi Xie,* Chunfeng Ma,* and Guangzhao Zhang



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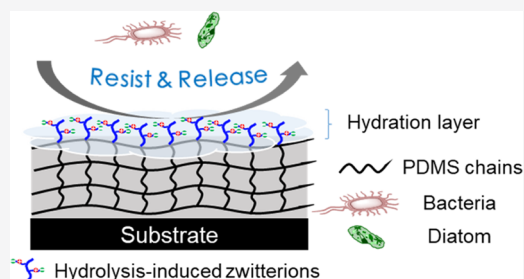


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Supporting Information

ABSTRACT: Silicone elastomer-based fouling release coatings have been gaining increased attention in marine antibiofouling. However, the lack of fouling resistance limits their application. Introducing a zwitterionic polymer into silicone enhances its fouling resistance, but their incompatibility makes this challenging. In this work, a silicone elastomer with zwitterionic pendant chains has been prepared by grafting a **telomer of tertiary carboxybetaine dodecafluoroheptyl ester ethyl acrylate (TCBF)** and 3-mercaptopropyltriethoxysilane to the bis-silanol-terminated poly(dimethylsiloxane) (PDMS). The fluorocarbon groups drive the telomer onto the surface in the film formation process, while the TCBF groups hydrolyze and generate zwitterions on the surface, which is confirmed by **attenuated total reflection infrared spectra** analysis and water contact angle measurements. Bioassays using marine bacteria (*Pseudomonas* sp.) and diatoms (*Navicula incerta*) demonstrate that the antifouling efficacy is improved as the telomer content increases. The bacteria and diatom adhesion decreases by 95 and 81%, respectively, for the PDMS with 30 wt % telomer compared with the unmodified PDMS control. Meanwhile, the fouling release performance of PDMS is maintained with a **pseudobarnacle** removal strength of ~ 0.1 MPa. This work provides a facile way to fabricate efficient silicone-based antifouling coatings.



INTRODUCTION

Unwanted marine biofouling settlement, such as barnacles, algae, oysters, and mussels, often occurs on surfaces immersed in ocean,^{1–3} which causes detrimental effects on marine industries and activities.⁴ The traditional approach to inhibit marine biofouling relies on biocide-based coatings.⁵ However, these highly toxic biocides lead to severe environmental damage. Thus, much attention has been focused on finding environmentally friendly antifouling coatings, such as zwitterionic polymers,^{6,7} silicone-based fouling release coatings,^{8,9} amphiphilic copolymers,^{10,11} degradable polymers,^{12,13} and liquid-infused slippery surfaces.^{14,15}

Silicone elastomers that can weaken the adhesion strength of the fouling organisms have been developed as an alternative to fouling control. In addition, they have the ability to reduce drag. The fouling release ability of these silicone elastomers, which usually consist of poly(dimethylsiloxane) (PDMS), stems mainly from their low surface energy (SE) and low elastic modulus.¹⁶ However, they cannot prevent the accumulation of the fouling organism under static conditions, because the fouling release requires a strong water flow, thus limiting their practical application.^{4,16} In recent years, the chemical grafting of antifouling moieties, such as amphiphile,^{10,17,18} zwitterion,^{19,20} and quaternary ammonium salt,^{21,22} has been explored to investigate whether the antifouling performance of silicone can be improved. Zwitterionic polymers can effectively inhibit nonspecific protein adsorption, bacterial adhesion, and biofilm formation

with advantages of easy functionality and design flexibility.^{23–25} In a typical method, the surface grafting of the zwitterions is undertaken to increase the surface antifouling moieties without compromising the bulk property. For instance, Jiang et al. grafted carboxybetaine polymer brushes onto PDMS surfaces using surface-initiated atom transfer radical polymerization and showed that the coatings exhibit good antifouling performance.²⁶ Yet, surface grafting requires complex preparation and harsh reaction conditions, and these coatings are usually effective only in high graft density.^{27,28}

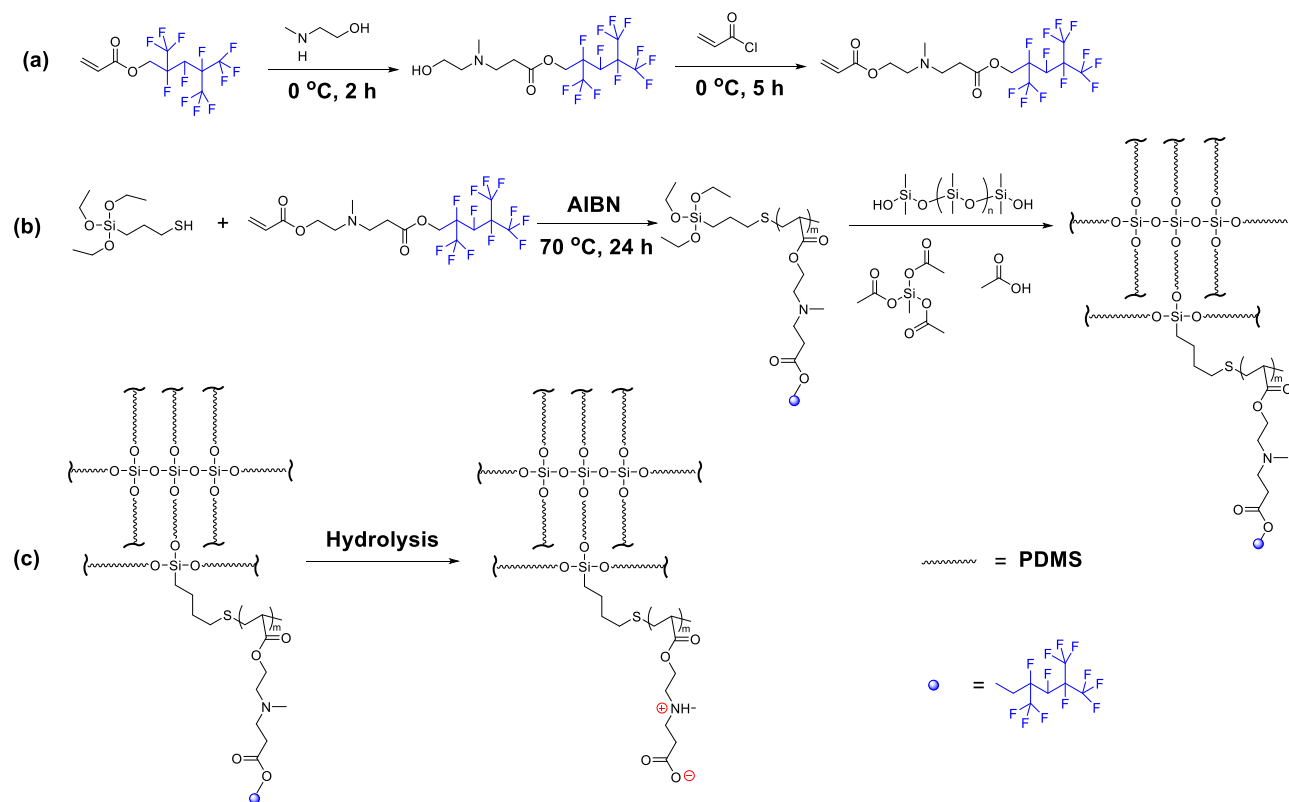
Another method to improve the fouling resistance of silicone is through the physical addition of antifouling moieties. Webster et al. prepared a series of poly(sulfobetaine methacrylate) triblock copolymers and incorporated them into a PDMS-polyurethane coating.²⁹ These coatings showed good fouling release efficiency toward bacteria and diatoms, but their performance may decline with time as the antifouling moiety reduces gradually. Ulbricht et al. introduced zwitterionic polymers into PDMS matrix, forming an interpenetrating polymer network, which showed low adsorption of proteins.³⁰

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Scheme 1. Synthetic Routes of TCBF (a), tTCBF and PDMS tTCBF-*x* (b), and the Hydrolysis of PDMS tTCBF-*x* (c)

A layer-by-layer assembly method was also used to modify silicone elastomers with zwitterionic polymers for the reduction of biofilm formation.³¹ Chemically grafting in bulk can solve the problem of release; however, PDMS has a low SE, while zwitterionic polymers have high SE, which causes the latter to be trapped in the bulk, so they cannot exert their fouling resistance ability. Another drawback is that zwitterions increase the hydrophilicity of silicone and affect the swelling ratio as well as the modulus of the coatings, which undermine the fouling release ability. Moreover, the incompatibility between highly polar zwitterionic materials and other nonpolar building blocks leads to phase segregation that may compromise the mechanical properties.

In the present study, a novel oligomer of zwitterionic monomer-tertiary carboxybetaine dodecafluoroheptyl ester ethyl acrylate (TCBF) and 3-mercaptopropyltriethoxysilane was synthesized by telomerization, and the telomer was chemically grafted to bis-silanol PDMS by a condensation reaction. The fluorocarbon can drive the telomer to migrate onto the surface before the coating is completely dry. TCBF is expected to undergo hydrolysis to produce self-generating zwitterions that enhance the fouling resistance ability during static conditions. Meanwhile, the rest of the TCBF telomer in bulk still retains its hydrophobicity, and it can reduce the swelling of PDMS coatings and maintain the fouling release performance. Within this context, the wettability, modulus, and resistance to typical marine organisms of the coatings were studied with the aim of developing eco-friendly and efficient antifouling coatings.

EXPERIMENTAL SECTION

Materials. 3-Mercaptopropyltriethoxysilane (MPTS), methyltriethoxysilane (MeAc), methylaminoethanol, acryloyl chloride, acetic

acid (AA), and triethylamine (TEA) were purchased from Aladdin and used as received. Dodecafluoroheptyl acrylate (DFA) was purchased from Zhengzhou JACS Chem Product Co., Ltd. and used as received. 2,2-Azobisisobutyronitrile (AIBN, Aladdin) was recrystallized from methanol prior to use. Bis-silanol-terminated PDMS (DMS-S31, $M_w = 26\,000$ g/mol) was purchased from Gelest. Tetrahydrofuran (THF) from Sinopharm was stirred with CaH₂ and distilled prior to use. Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2003).

Synthesis of Tertiary Carboxybetaine Dodecafluoroheptyl Ester Ethyl Acrylate (TCBF). The synthetic route of TCBF is shown in Scheme 1. Methylaminoethanol (1.94 g) was added to 10.0 g of DFA dropwise under magnetic stirring at 0 °C in an argon atmosphere. After stirring for 3 h, 45 mL of THF and 2.62 g of TEA were added to the reaction contents, and then 2.69 g of acryloyl chloride was added dropwise at 0 °C. After stirring for 3 h, the mixture was filtered and THF was evaporated under rotary evaporation. The resulting liquid was dissolved using ethyl acetate and then extracted by a saturated NaCl solution. The ethyl acetate layer was dried using Na₂SO₄, filtered, and concentrated to give a transparent oily product. ¹H nuclear magnetic resonance (NMR) (600 MHz, CDCl₃, ppm, Figure S1): 6.40 (CH₂CH-), 6.14 (CH₂CHCOO-), 5.83 (CH₂CH-), 5.47 (-CFCF₃CFHCFCF₃-), 4.62 (-NCH₃CH₂CH₂-), 4.24 (-COOCH₂CFCF₃-), 2.80 (-COOCH₂CH₂-), 2.72 (-CH₂CH₂NCH₃-), 2.56 (-CH₂CH₂COO-), 2.31 (-CH₂NCH₃CH₂-).

Synthesis of the Telomer (tTCBF). The telomerization of MPTS and TCBF was conducted following a procedure stated elsewhere.^{10,32} Typically, MPTS (0.92 g), TCBF (10.00 g), and AIBN (0.02 g) were dissolved in 20 mL of THF and introduced to a Schlenk tube with a magnetic stirring bar. The solution was degassed by three freeze–evacuate–thaw pump cycles. The tube was then filled with argon, and the reaction proceeded at 70 °C for 24 h. The resultant polymer was precipitated in hexane and dried under vacuum at 40 °C for 12 h. ¹H NMR (600 MHz, CDCl₃, ppm, Figure 1): 5.50 (-CFCF₃CFHCFCF₃-), 4.62 (-NCH₃CH₂CH₂-), 4.13 (-COOCH₂CFCF₃-), 3.80 (CH₃CH₂OSi-), 2.54

($-\text{CH}_2\text{CH}_2\text{COO}-$), 2.27 ($-\text{CH}_2\text{NCH}_3\text{CH}_2-$), 1.22 ($\text{CH}_3\text{CH}_2\text{OSi}-$), 0.70 ($-\text{SiCH}_2\text{CH}_2\text{CH}_2-$).

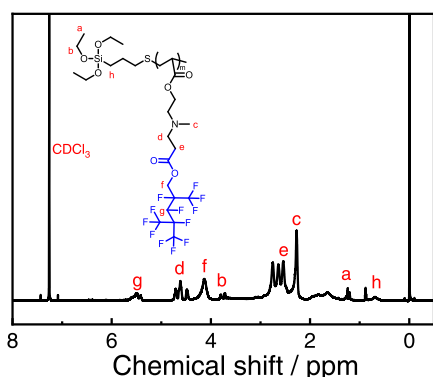


Figure 1. ^1H NMR spectra of tTCBF.

Preparation of PDMS-Based Coating with Grafted Telomer.

The coating was prepared using a condensation reaction (Scheme 1). Typically, bis-silanol-terminated PDMS (10.0 g), MeAc (1.50 g), AA (1.50 g), and tTCBF (1.28 g) were dissolved in 100 mL of THF and stirred for 1 h. The solution was cast onto silicon wafer (for the antibacterial assays, antidiatom tests, and surface modulus tests), glass slide (for pseudobarnacle tests and surface wettability tests), and glass fiber-reinforced epoxy panel (for water absorption test and adhesion test). The thickness of the dry samples was $\sim 200\ \mu\text{m}$. The coating was cured at room temperature for 5 days and designated as PDMS tTCBF- x , where x refers to the mass percentage of the telomer in the coating.

Characterization. *Proton Nuclear Magnetic Resonance Spectroscopy (^1H NMR).* ^1H NMR analyses were performed on a Bruker AV600 NMR spectrometer using CDCl_3 as the solvent and tetramethylsilane (TMS) as the interior label.

Attenuated Total Reflection Infrared (ATR-IR) Spectroscopy and Fourier Transform Infrared (FTIR) Spectroscopy. The nonhydrolyzed films were recorded using a Bruker VERTEX 70 infrared spectrometer with a ZnSe crystal. The resolution was $4\ \text{cm}^{-1}$, and the accumulation was 32 scans. After immersion in ASW for 3 days, the hydrolyzed films were studied using the same method. FTIR spectra were also recorded on a Bruker VERTEX 70 spectrometer using the KBr disk method in the scanning range of 4000 to $400\ \text{cm}^{-1}$ and at $4\ \text{cm}^{-1}$ resolution.

Extraction Studies. To confirm the chemical grafting of the telomer, an extraction test was done on a typical sample PDMS tTCBF-20. The coating on glass slide with a known weight was immersed in THF and shaken for 0.5 h. Then, the solution was filtered and THF was removed under reduced pressure. The leachate with a certain amount of N,N -dimethylformamide (DMF, as internal reference) was tested with ^1H NMR, and the mass of tTCBF was calculated.

Water Contact Angle (WCA) and Surface Energy (SE) Measurements. The WCA was determined using a theta optical tensiometer (Attension, Biolin Scientific) at $25\ ^\circ\text{C}$ by dropping $3\ \mu\text{L}$ of deionized water droplets on the coating surface using the sessile method. After immersion in ASW for different time periods, the film was rinsed with deionized water and dried with nitrogen flow and the WCA was measured using the same method. Five regions were tested with each film to obtain the average value. The SE was calculated from the WCA and the methylene iodide contact angle using the Owens–Wendt–Rabel–Kaelble method.³³

Atomic Force Microscopy (AFM) Measurements. The surface elastic modulus measurements were conducted with an XE-100 AFM (Park Systems) by applying Hertzian contact theory, which correlates the AFM force signal (F) to the indentation depth of the cantilever tip (δ) and the elastic modulus (E) using the following equation³⁴

$$F = \frac{E}{1 - \nu^2} \frac{\tan \alpha}{\sqrt{2}} \delta^2$$

where ν is Poisson's ratio (0.5 for all coatings), α is the face angle of the four-sided pyramidal tip of the cantilever (22°), and δ is the indentation depth.

Water Absorption Test. The samples on an epoxy panel ($2 \times 2\ \text{cm}^2$) were prepared via solution casting. The weight of each epoxy panel (W_p) was recorded, and the initial weight (W_0) of the dried sample together with the slide was measured before it was immersed in ASW at $25\ ^\circ\text{C}$. After immersion, it was taken out of the ASW at different time intervals and rinsed with deionized water, and the surface water was drained using airlaid paper. After that, the coated slide was weighed and denoted as W_t . The water absorption was calculated by $(W_t - W_0)/(W_0 - W_p)$. Three parallel samples were measured for each coating.

Adhesion Test. The adhesion strength of PDMS tTCBF- x coatings on glass fiber epoxy resin was measured on a Posi Test AT-A Automatic tester according to ASTM D4541. The surface of the coatings was roughed using 60-grit abrasive papers. An aluminum dolly with a diameter of 20 mm was attached to the coating using epoxy adhesive (Araldite Rapid Epoxy Adhesive, Huntsman) and cured at room temperature for 3 days. The pull rate was $0.2\ \text{MPa}\cdot\text{s}^{-1}$, and five different regions were measured to obtain an average value.

Pseudobarnacle Test. The fouling release test was characterized by the removal of pseudobarnacles according to the ASTM D5618 (2011). The samples were first immersed in ASW for 3 days and then dried by the nitrogen flow. Five pseudobarnacles made of a cylindrical aluminum stud (10 mm in height and 10 mm in diameter) were attached to the coating surface with adhesive (Araldite Rapid Epoxy Adhesive, Huntsman, thickness of $50\ \mu\text{m}$). The adhesive was cured for 3 days under ambient conditions. The removal strength was tested using a force gauge (SUNDOO SH-500).

Antibacterial Assays. The marine bacterium *Pseudomonas* sp. was used to examine the antibacterial performance. The film was immersed in 1 mL bacterial suspension of ASW (10^8 cells/mL) at $30\ ^\circ\text{C}$ for 5 h. Subsequently, the film was washed gently with ASW to remove the unattached bacteria and was stained with the LIVE/DEAD BacLight Bacterial Viability Kit for 0.5 h. The adhered cells were examined under a fluorescence microscope (Scope A1, Zeiss). The number of bacteria was determined from three random fields of view on triplicate slides. Data were normalized to those of a control sample.

Antidiatom Test. The diatom *Navicula incerta* was employed to investigate the antidiatom property of the coating. Guillard's F/2 culture medium was used to cultivate *N. incerta* in a plant growth chamber (Bluepard) at $23\ ^\circ\text{C}$ with a 12:12 h light–dark cycle. The polymers were coated on the silicon wafer, and the wafer was then put inside the wells of a 24-multiwell plate. One milliliter of diatom suspension was added to each well and incubated for 24 h. The glass slides were washed gently with ASW to remove the unattached diatoms. Counting was performed using microscopy (Scope A1, Zeiss), and 10 random fields of view on triplicate slides were selected to quantify the diatom density.

RESULTS AND DISCUSSION

The synthetic routes, the typical ^1H NMR spectra of TCBF, and the telomer tTCBF are shown in Scheme 1, Figure S1, and Figure 1, respectively. The peaks at 1.22 and 3.80 ppm are attributed to the ethoxy group in silane, the signals of the fluorocarbon group can be observed at 4.13 and 5.50 ppm, and the peak at 2.27 ppm can be assigned to the CH_3 linked to the nitrogen atom. These results clearly indicate the successful synthesis of the telomer. The molar mass was calculated to be $\sim 2800\ \text{g/mol}$ according to the ^1H NMR. The telomer was subsequently mixed with bis-silanol-terminated PDMS at room temperature in the presence of a cross-linking agent MeAc and a catalyst AA, generating a modified silicone coating. Coatings

with three different contents of tTCBF (10, 20, and 30%) were prepared in this work, and the coating formulations are shown in Table 1. To calculate the proportion of the telomers cross-

Table 1. Composition of PDMS tTCBF-*x* Coatings

sample	PDMS matrix (g)	TCBF telomer (g)	MeAc (g)	AA (g)
PDMS	10	0	1.5	1.5
PDMS tTCBF-10	10	1.28	1.5	1.5
PDMS tTCBF-20	10	2.87	1.5	1.5
PDMS tTCBF-30	10	3.94	1.5	1.5

linked into the elastomers, extraction studies were done using PDMS tTCBF-20. ^1H NMR showed that more than 95% of the telomer was cross-linked into the elastomers (Figure S2), indicating the high grafting efficiency.

The hydrolysis of grafted zwitterionic precursors with the aid of ASW was studied with the WCA and IR. Figure 2 shows the

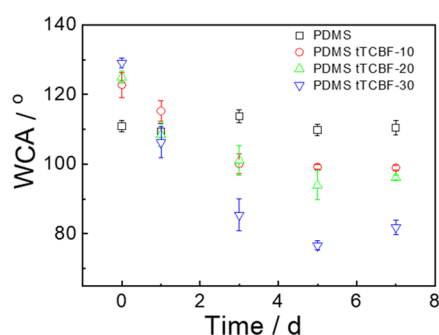


Figure 2. Time dependence of the WCA of coating surfaces after immersion in ASW.

time dependence of the WCA on the coating surfaces. The WCA of PDMS is $\sim 110^\circ$, which indicates its high hydrophobicity. This is consistent with the data presented in previous reports.^{35–37} The PDMS tTCBF-*x* coatings have more hydrophobic surfaces with WCAs $>120^\circ$, and these WCAs increase with the telomer contents because of the long fluorocarbon segment in tTCBF that drives the migration on account of its low SE and incompatibility with silicone. After immersion in ASW, the WCA of PDMS changes slightly, given that the cross-linked structure of PDMS makes the surface stable. The WCAs of PDMS tTCBF-*x* coatings decrease significantly, which indicates the presence of carboxylate groups produced through the hydrolysis of the TCBF groups. In particular, the higher the tTCBF content, the faster the WCA declines (reaching $\sim 80^\circ$ for PDMS tTCBF-30) as a great number of zwitterions are yielded during the hydrolysis process as the tTCBF content increases.

After hydrolysis in ASW for 3 days, the samples were chosen in the following characterization, given that the WCA slightly changed and leveled to a plateau after 3 days of immersion. The hydrolysis of the telomer tTCBF was first studied by FTIR spectroscopy. As shown in Figure S4, the peak at 1235 cm^{-1} assigned to the stretching vibrations of C–F bonds disappeared after hydrolysis, indicating the hydrolysis of tTCBF.⁹ Moreover, the new peak at 1540 cm^{-1} was assigned to COO^- groups, indicating the formation of zwitterions.³⁸ Figure 3 shows the ATR-IR spectra of the PDMS tTCBF-*x* coatings before and after hydrolysis. The peaks in the spectra of PDMS do not change, showing that the chemical structure is

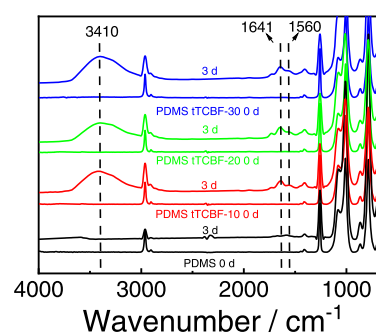


Figure 3. ATR-IR spectra of PDMS tTCBF-*x* coatings before and after immersion in ASW for 3 days.

stable in ASW. Regarding the PDMS tTCBF-*x* coatings, a new band attributed to COO^- can be observed at 1560 cm^{-1} , indicating that the dodecafluoroheptyl ester group is hydrolyzed into negatively charged carboxyl group and the tertiary amines in the polymer were protonated.²⁵ The strong absorption peak at 1641 cm^{-1} is attributed to the bending vibration of water molecule, which is somewhat overlapped with the characteristic carboxyl peak around this wavenumber.^{39–41} Furthermore, the band at 3410 cm^{-1} is attributed to OH from the hydration of the zwitterions. After normalization with the absorption peak at 2960 cm^{-1} , the integrated intensity of the COO^- region at 1560 cm^{-1} for PDMS tTCBF-30 and 20 are 4.3 and 2.6 times that of PDMS tTCBF-10, respectively, indicating that more zwitterions are generated on the surface. While the PDMS tTCBF-*x* coatings were immersed in ASW, they have swelling because of the presence of zwitterions. As shown in Figure S4, there is a positive correlation between the tTCBF content and the water absorption, which reached a plateau after immersion for 7 days. It should also be noted that the water absorption is not high, even for the coating containing 30 wt % of tTCBF because most zwitterions are on the surface, which prevents problems derived with serious water absorption.^{42,43}

The SE and surface elastic modulus of the PDMS tTCBF-*x* coatings after immersion in ASW for 3 days were also studied, as shown in Figure 4. PDMS has an SE of $22\text{ mJ}\cdot\text{m}^{-2}$. The SE increased with the tTCBF-*x* content, ranging from 25 to $32\text{ mJ}\cdot\text{m}^{-2}$. This increase correlates with the WCA. The greater the change in WCA values, the greater the change in the SE of a coating.^{44,45} The surface modulus is also an important factor of the fouling release performance, where a lower modulus is favorable for fouling release.^{46,47} Figures 4b and S5 show the surface elastic modulus and the force-indentation curve of PDMS tTCBF-*x* coatings after immersion. PDMS has a modulus of $\sim 0.9\text{ MPa}$, but PDMS tTCBF-*x* coatings exhibit an even lower modulus ($\sim 0.6\text{ MPa}$). The reasons are twofold. First, the introduction of zwitterionic polymer chains reduces the interactions between the polysiloxane chains. Second, the water molecules absorbed can also reduce the inter/intramolecular interactions of polymer chains. In general, the coatings have relatively low SE and modulus and are expected to possess good fouling release ability.

As the PDMS-based coatings often have good fouling release properties, marine organisms, such as barnacles, that are attached to the surface can be released easily with shear force. Pseudobarnacle removal tests were conducted to verify whether the modified PDMS coatings still possessed their fouling release ability after hydrolysis in ASW for 3 days

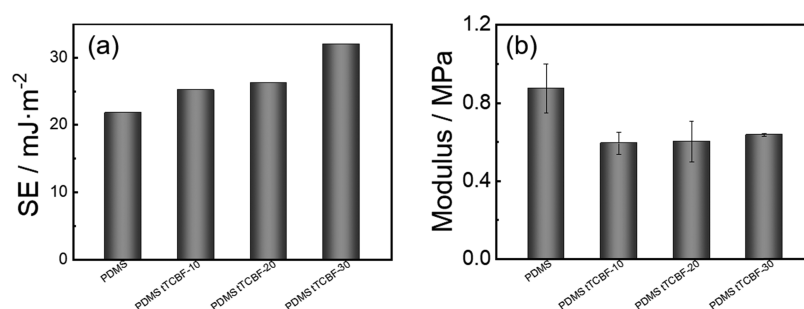


Figure 4. Surface energy (SE) (a) and surface modulus (b) of PDMS tTCBF- x coatings after immersion in ASW for 3 days.

(Figure 5). PDMS demonstrates excellent fouling release performance with a low removal strength of ~ 0.05 MPa.

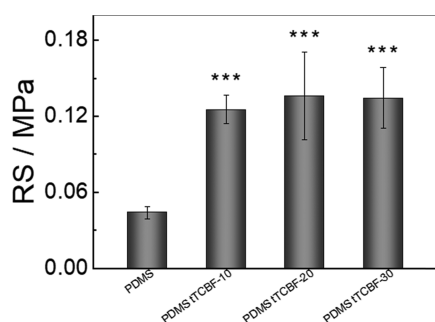


Figure 5. Removal strength (RS) of pseudobarnacles adhered to PDMS tTCBF- x coatings after immersion in ASW for 3 days. (***) denotes significant difference at $p < 0.001$, compared to the results of PDMS and determined by the one-way ANOVA test.

PDMS tTCBF- x coatings exhibit a low removal strength (~ 0.12 MPa) with a significant difference with PDMS ($p < 0.001$), but this is slightly higher than that of unmodified PDMS, indicating that the fouling release ability declined slightly. It is known that the fouling release ability is related to the SE and elastic modulus,⁴⁸ and as discussed above, although the elastic modulus decreases slightly after hydrolysis in ASW, the SE increases markedly, which is most likely responsible for this decrease of fouling release ability. Nevertheless, PDMS tTCBF- x coatings still have excellent fouling release ability according to the removal strength values observed in the current study and in previous publications.^{35,49,50} Ideally, the silicone elastomer coatings should have both good adhesion to the substrate and fouling release properties. However, it is very difficult to achieve the desired result due to the low surface energy of PDMS. Generally, the adhesion strength of PDMS

elastomer is about 0.5 MPa, but can increase to 1.5 MPa using a tie layer.⁵¹ The matrix of the PDMS tTCBF- x coatings is mainly based on PDMS elastomer, and the adhesion strength of PDMS tTCBF coatings decreased from 0.5 to 0.3 MPa with the content of tTCBF (Figure S6). Thus, the system should combine the tie layer for future applications.

The antifouling performance of the PDMS tTCBF- x coatings was evaluated using two model marine organisms: marine bacteria *Pseudomonas* sp. and diatom *N. incerta*. Figure 6 shows the fluorescence microscopy images of the antibacterial assays and the relative bacterial settlement (RBS) on the surfaces. Zwitterionic polymers are fouling-resistant polymers that prevent the adsorption of bacteria instead of killing them.³⁹ A large number of live bacteria were observed on the surfaces of the blank silicon wafer as well as the PDMS before and after hydrolysis ($\sim 99\%$ bacterial adhesion), indicating that the fouling pressure was high and that PDMS is highly subjected to bacterial settlement. Without prehydrolysis in ASW, the PDMS tTCBF- x coatings showed a similar bacterial settlement (Figure 6a) to PDMS, which resulted from the hydrophobic surfaces without zwitterions. When the samples were prehydrolyzed in ASW for 3 days, the PDMS tTCBF- x coatings mitigated the bacterial settlement compared with the blank and PDMS surfaces, and the antibacterial performance was improved by increasing the tTCBF content. Compared with the blank control, the reduction of RBS for PDMS tTCBF-10, -20, and -30 was 38, 79, and 95%, respectively, because of the generated zwitterions on the surfaces, which effectively resisted the settlement of bacteria. The results also revealed that the strategy of hydrolysis-induced zwitterions was competent in preventing the settlement of bacteria.

Diatoms are another representative fouling organisms that settle easily on submerged surfaces. Figure 7 shows the optical images of the diatom *N. incerta* adhered on the PDMS tTCBF-

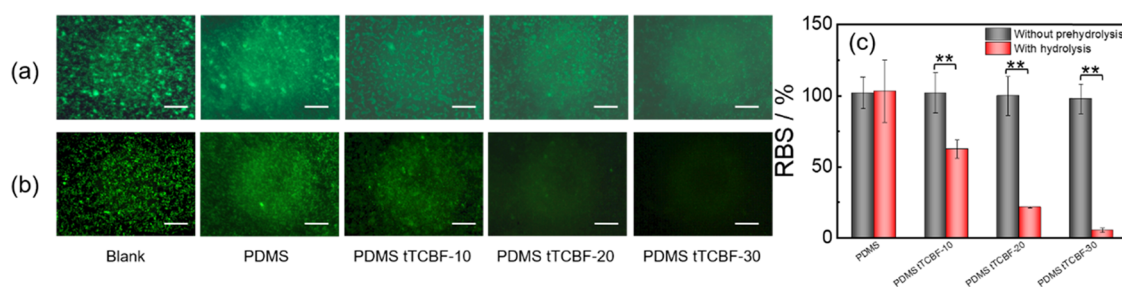


Figure 6. Fluorescence microscopy images of *Pseudomonas* sp. settled to PDMS tTCBF- x coatings without (a) and with (b) immersion in ASW for 3 days, and relative bacterial settlement (RBS) (c). The scale bar is $50\ \mu\text{m}$. (**) denotes significant difference at $p < 0.01$ determined by the one-way ANOVA test.

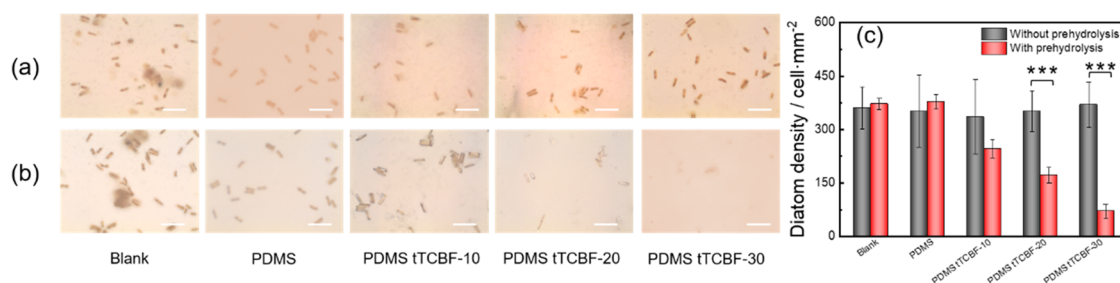


Figure 7. Optical microscopy images of the diatom *N. incerta* adhered on PDMS tTCBF-*x* coatings without (a) and with (b) prehydrolysis in ASW for 3 days, and the diatom density (c). The scale bar is 50 μm . (***) denotes significant difference at $p < 0.001$ determined by the one-way ANOVA test.

x coatings and the diatom density. Compared with the blank silicon wafer control, a greater density of diatoms ($\sim 370 \text{ cell}\cdot\text{mm}^{-2}$) can be observed on the surfaces of PDMS with and without prehydrolysis. Diatoms can adhere easily to PDMS since PDMS materials are hydrophobic and possess poor ability to inhibit the adhesion of marine slime (which consists mainly of diatoms and bacteria).²⁹ Without prehydrolysis, the PDMS tTCBF-*x* coatings exhibited a similar diatom density to PDMS because no zwitterions were generated on the surfaces. After prehydrolysis, the PDMS tTCBF-*x* coatings show a lower adhesion, and the antidiatom performance was also improved by increasing the tTCBF content. This is consistent with the results of the antibacterial assays. The densities of diatoms settled on PDMS tTCBF-10, -20, and -30 are ~ 245 , ~ 170 , and $\sim 70 \text{ cell}\cdot\text{mm}^{-2}$, respectively, which indicates that PDMS tTCBF-*x* coatings with a certain number of zwitterions can effectively resist diatom adhesion. Both the antibacterial and antidiatom results demonstrate that PDMS tTCBF-20 and 30 coatings can effectively generate hydrolysis-induced zwitterionic surfaces to inhibit the marine biofouling.

CONCLUSIONS

In this study, a self-generating zwitterionic PDMS coating was prepared using a simple hydrolysis-inducing method. The tTCBF telomer that functioned as a zwitterionic precursor was blended with bis-silanol-terminated PDMS and self-stratified to the surface of the coating during film formation. It was observed that PDMS tTCBF-*x* coatings generated zwitterionic moieties on the coating surface after hydrolysis in ASW, while maintaining relatively low water adsorption of the bulk, and the modified PDMS-based coatings still exhibited a good fouling release performance. Moreover, the coatings containing 20 and 30 wt % of tTCBF exhibited a good fouling-resistant performance toward both bacteria (*Pseudomonas* sp.) and diatom (*N. incerta*) adhesion. The coating is expected to be applied in marine antibiofouling.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.1c00984>.

¹H NMR spectra of TCBF and the extract leachate of PDMS tTCBF-20, FTIR spectra of tTCBF before and after 3 days of hydrolysis, water absorption of PDMS tTCBF-*x* coatings versus immersion time, force-indentation curve of PDMS tTCBF-*x* coatings by AFM, and the adhesion strength (AS) of PDMS tTCBF-*x* coatings (PDF)

AUTHOR INFORMATION

Corresponding Authors

Qingyi Xie – Faculty of Materials Science and Engineering, South China University of Technology, Guangzhou 510640, P. R. China; Email: xieqy@scut.edu.cn

Chunfeng Ma – Faculty of Materials Science and Engineering, South China University of Technology, Guangzhou 510640, P. R. China; orcid.org/0000-0002-1649-723X; Email: msmcf@scut.edu.cn

Authors

Peng Hu – Faculty of Materials Science and Engineering, South China University of Technology, Guangzhou 510640, P. R. China

Haohang Zeng – Faculty of Materials Science and Engineering, South China University of Technology, Guangzhou 510640, P. R. China

Huan Zhou – China Ship Development and Design Center, Wuhan 430064, P. R. China

Cong Zhang – China Ship Development and Design Center, Wuhan 430064, P. R. China

Guangzhao Zhang – Faculty of Materials Science and Engineering, South China University of Technology, Guangzhou 510640, P. R. China

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.langmuir.1c00984>

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